
Lab Seminar:

Quantum Deep Reinforcement Learning

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Outline of presentation

I. Introduction to Quantum Computing

II. Quantum Machine Learning

III. Quantum Deep Reinforcement Learning

IV. My idea for the conference

Introduction to Quantum Computing

Classical Bits vs. Qubits

Classical Bit: 0 or 1

Qubit: is a linear combination of these 2 states.

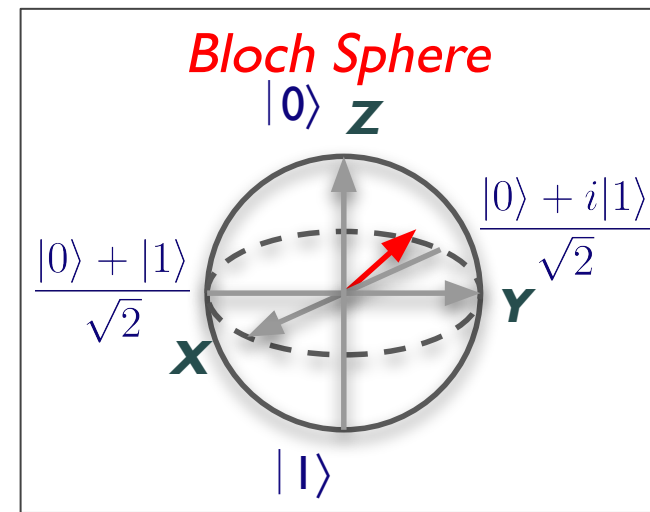
A **quantum state** of a qubit is:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

- Where: Amplitudes α and β are **complex numbers**, satisfying:
$$\alpha^2 + \beta^2 = 1$$

- Example:

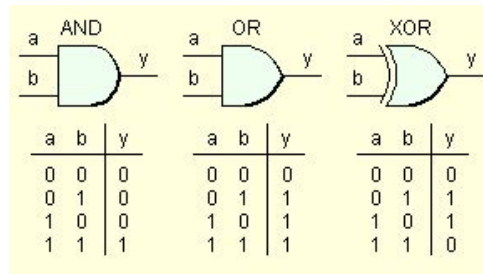
$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



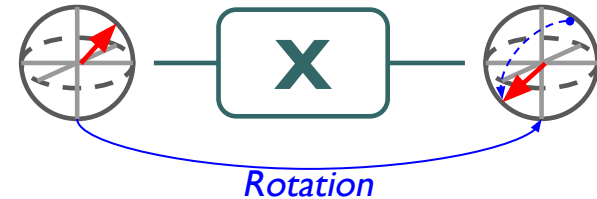
→ There are 3 ways to represent a qubit: (i) Dirac notation (ii) Matrix form (iii) Bloch sphere

Quantum gates

Classical Bit: Logic gates



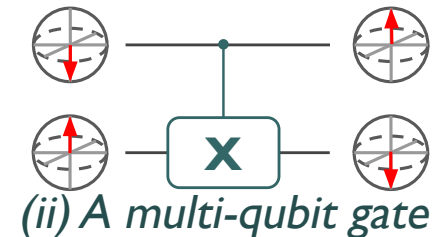
Qubit: Quantum gates



→ A **quantum gate** is an operation that changes the quantum state of a qubit, similar to how a logic gate changes the value of a bit in a classical computer.

Quantum gates:

- (i) Single-qubit gates
- (ii) Multi-qubit gates



Quantum gates are mathematically represented by **matrices**:

$$\begin{array}{ccccc}
 \boxed{X} & \boxed{Y} & \boxed{Z} & \boxed{H} & \text{CNOT} \\
 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} & \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \\
 \text{(a)} & \text{(b)} & \text{(c)} & \text{(d)} & \text{(e)}
 \end{array}$$

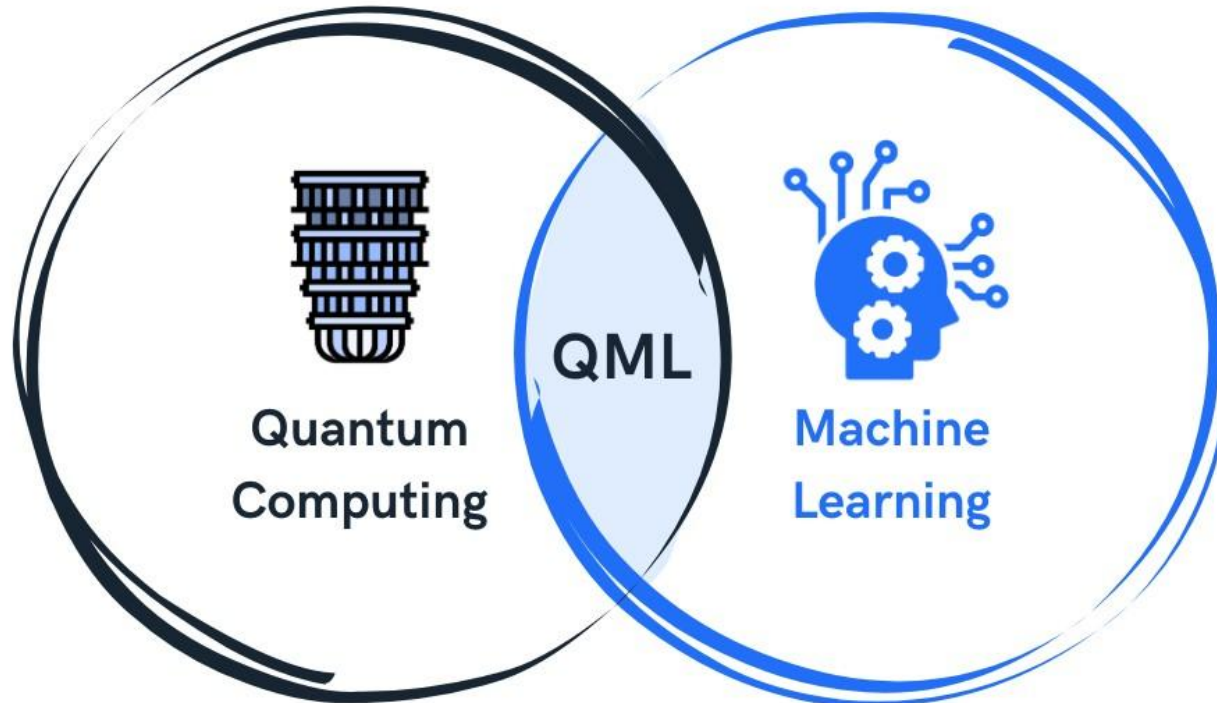
Quantum computation can be understood as computation through matrix operations in linear algebra.

Quantum Machine Learning

Machine Learning meets Quantum Computing

Quantum machine learning (QML) is a field that combines ideas from Quantum Computing and Machine Learning to create new algorithms that can run on quantum computers.

Motivation: The motivation of QML is to use Quantum Computing to make Machine Learning faster and more scalable by leveraging quantum properties.

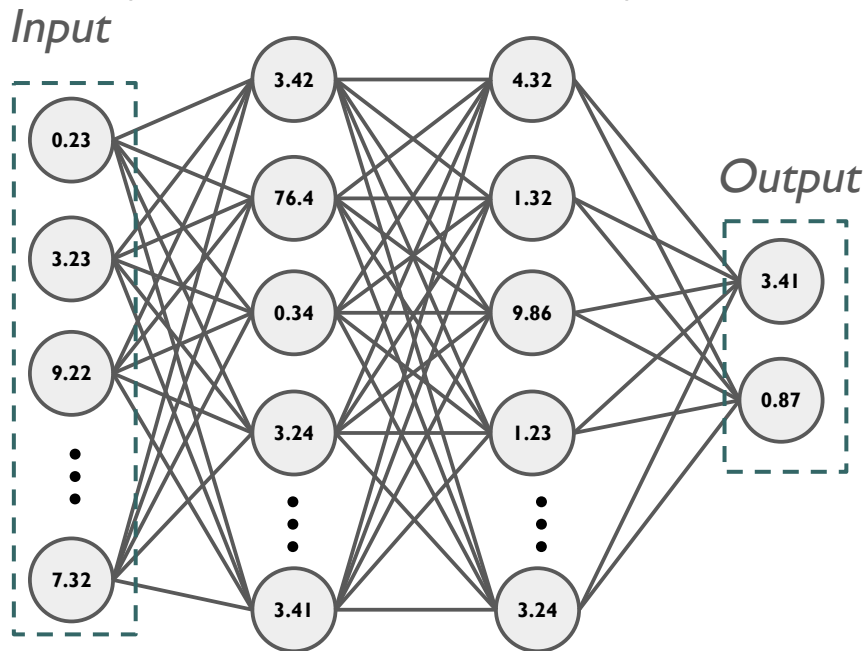


Machine Learning vs. Quantum Machine Learning

Machine Learning:

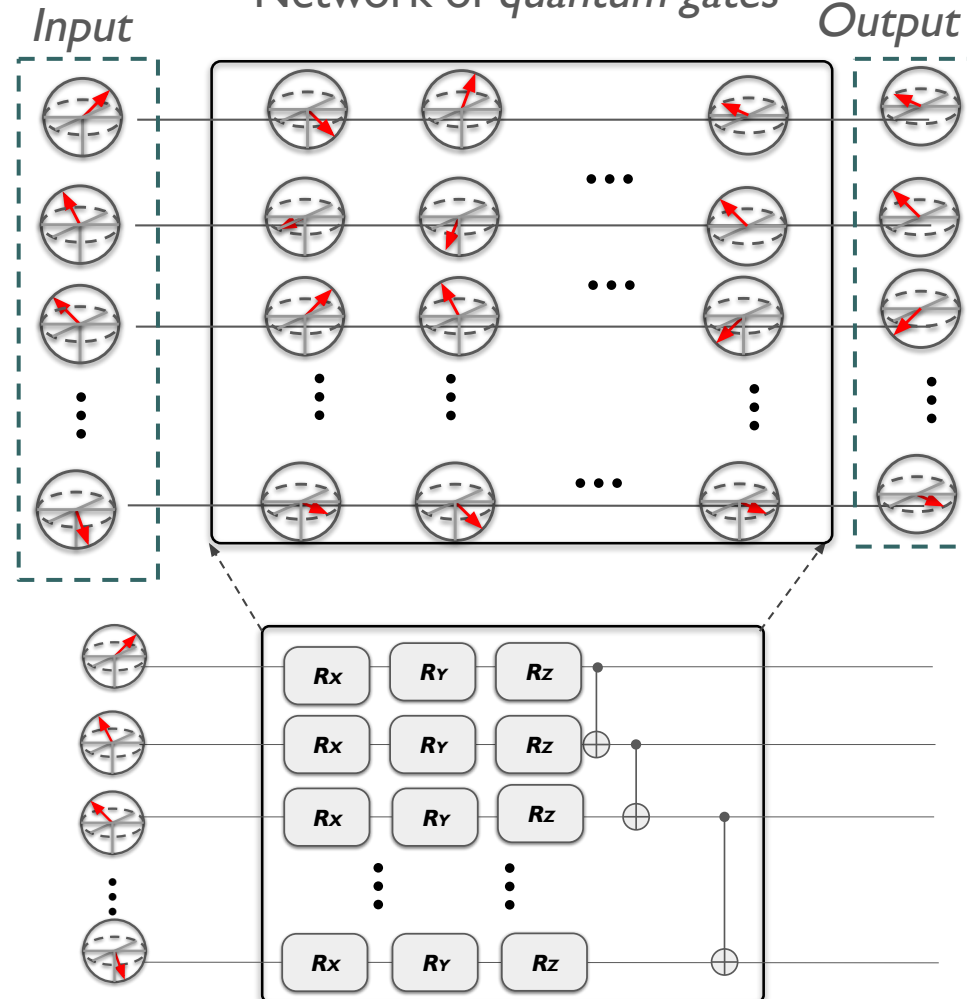
Deep Neural Networks

(Network of neural nodes)



Quantum Machine Learning: Variational Quantum Circuits (VQC)

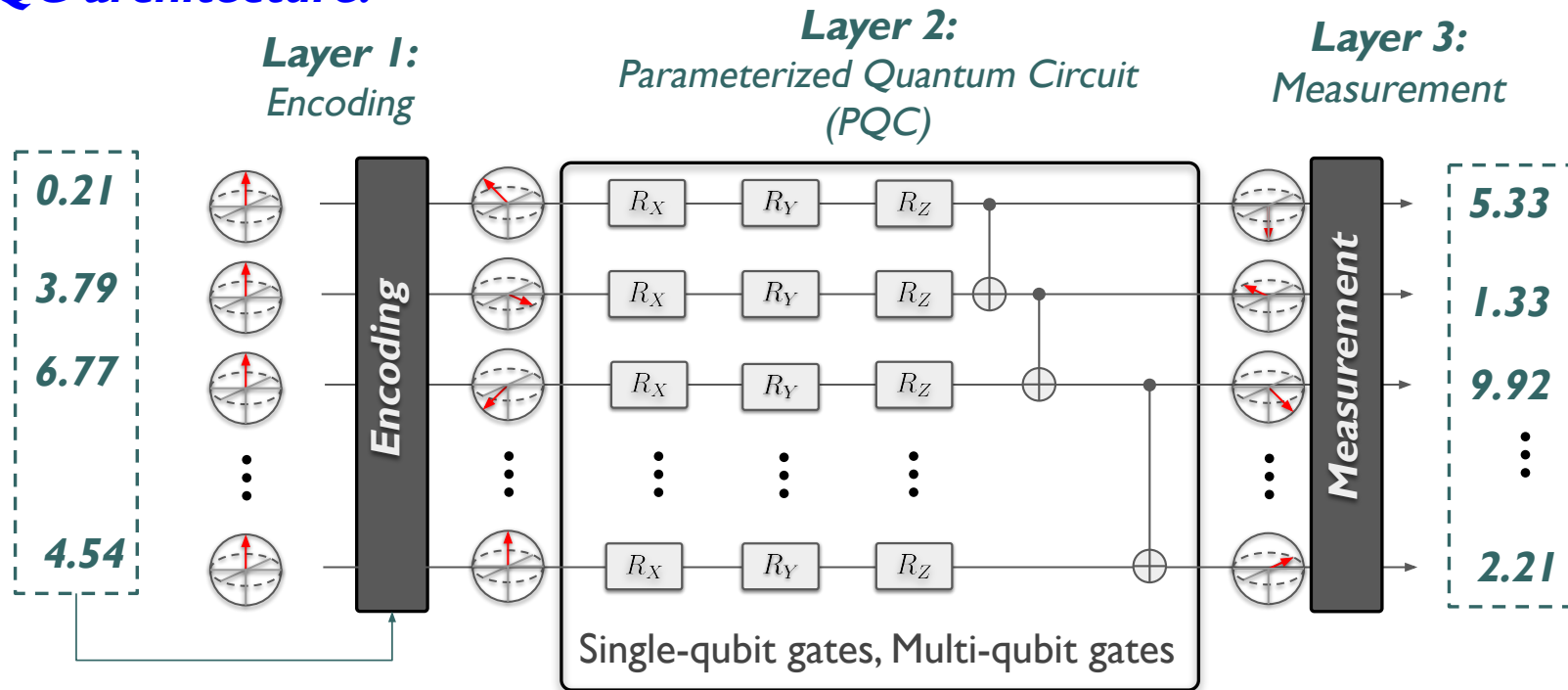
Network of quantum gates



Quantum ML: Variational Quantum Circuits

A **Variational Quantum Circuit (VQC)** is a parameterized quantum circuit with trainable parameters, widely used in QML algorithms.

VQC architecture:

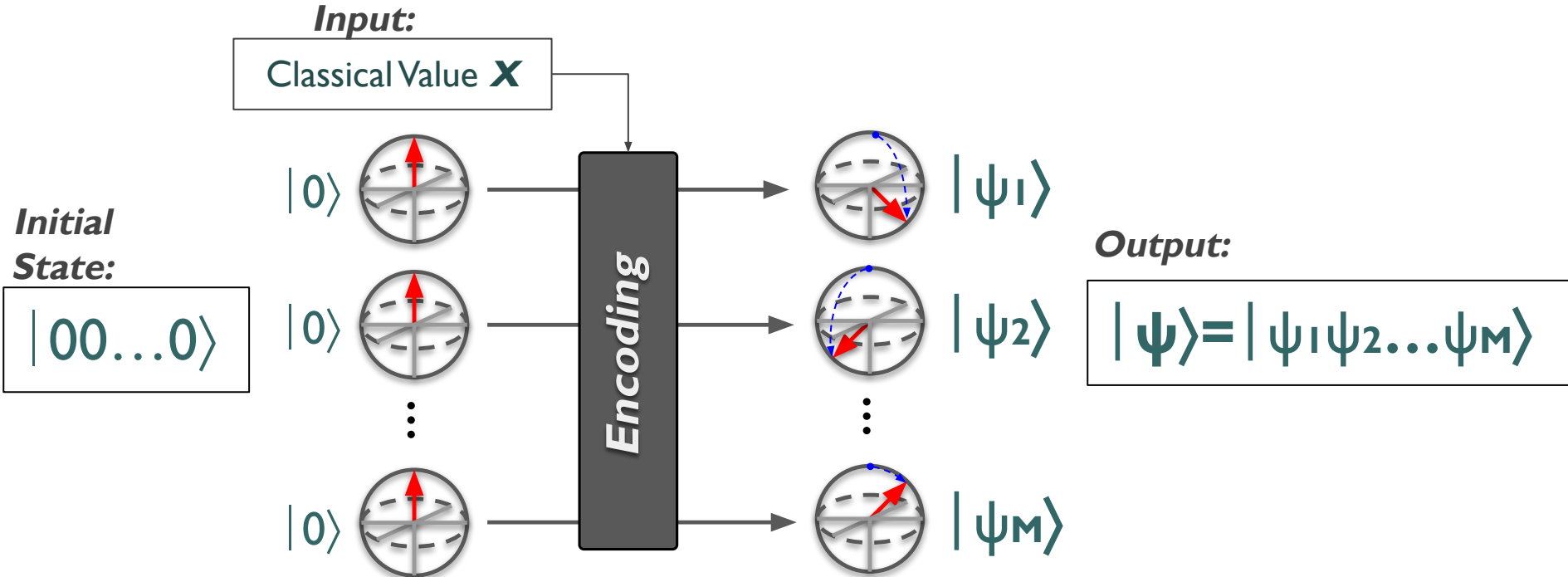


- **Layer 1-Encoding Layer:** Convert Classical value $\rightarrow$ Quantum value
- **Layer 2-PQC Layer:** Trainable Parameterized Quantum Circuit (PQC)
- **Layer 3-Measurement Layer:** Convert Quantum value $\rightarrow$ Classical value

Variational Quantum Circuits: Encoding Layer

Encoding Layer: There are 3 encoding techniques

1. *Basis Encoding*
2. *Angle Encoding*
3. *Amplitude Encoding*



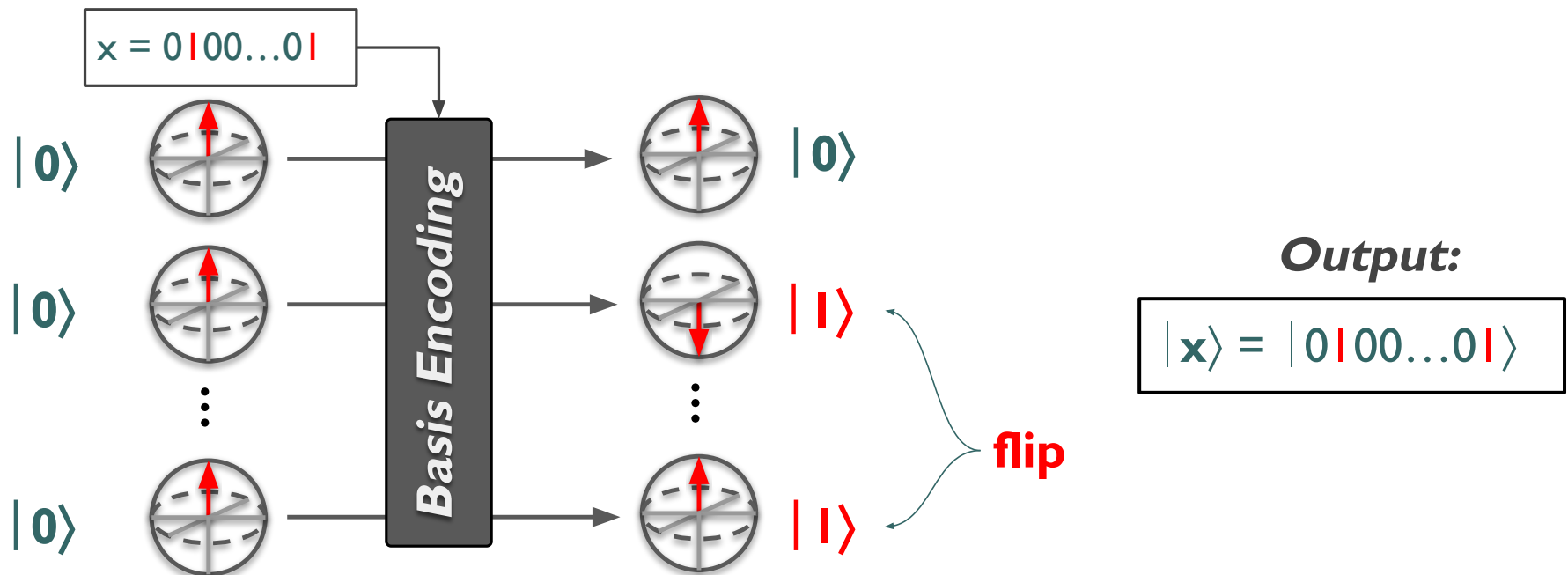
Encoding layer is converting $|0\rangle$ quantum states to new quantum states

Variational Quantum Circuits: Encoding Layer

Encoding Layer:

I. Basis Encoding: Represent classical data as a **binary string**, then map it directly to a **basis state**.

- Bit string $\mathbf{x} = x_1x_2\dots x_N \quad \forall x_i \in \{0,1\} \rightarrow$ Quantum value $|\mathbf{x}\rangle = |x_1x_2\dots x_N\rangle$
- Example: $x=5 \rightarrow x=101 \rightarrow$ Encoded $|\mathbf{x}\rangle = |101\rangle$

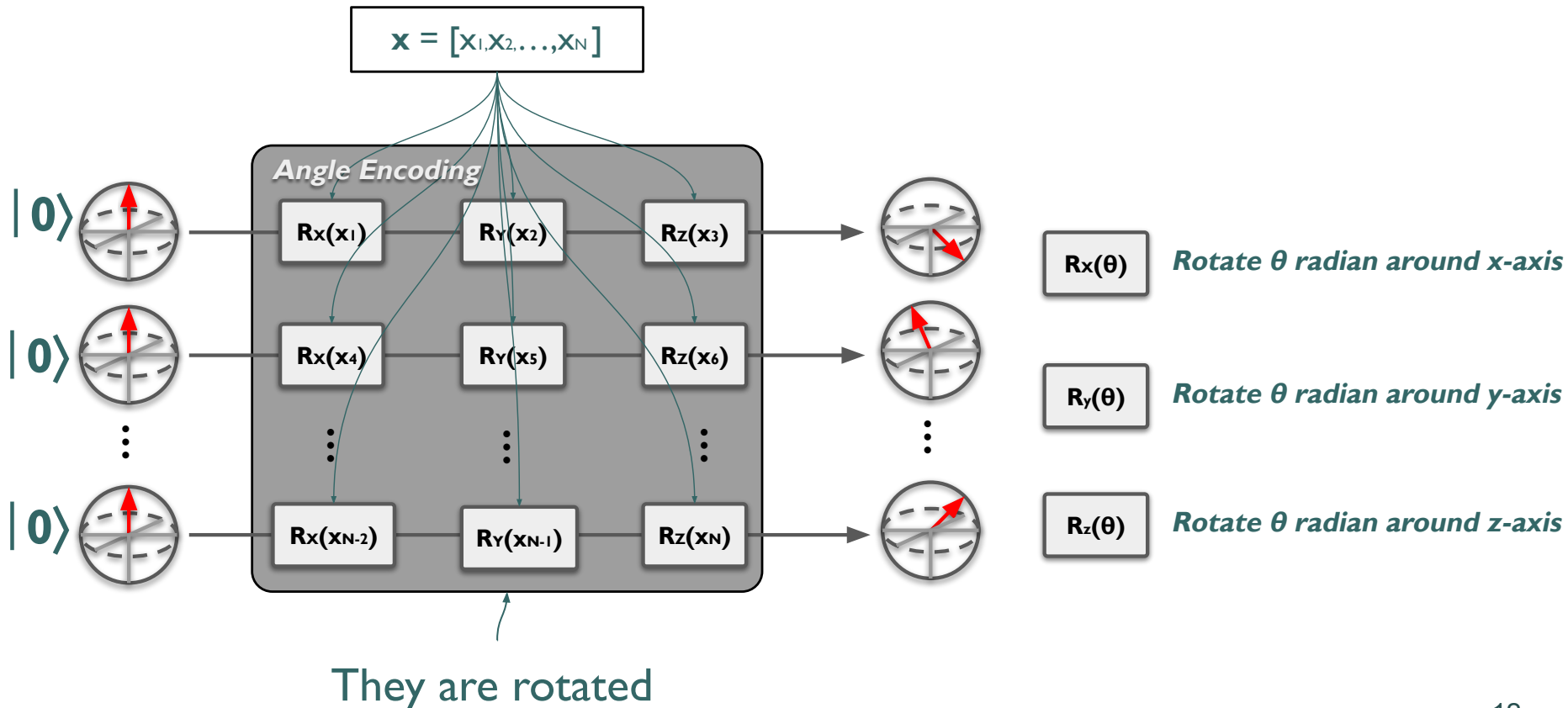


Variational Quantum Circuits: Encoding Layer

Encoding Layer:

2.Angle Encoding: Take each classical value x_n as a **rotation angle** applied to each qubit.

Given: Vector of classical values $\mathbf{x} = [x_1, x_2, \dots, x_N]$



Variational Quantum Circuits: Encoding Layer

Encoding Layer:

3. Amplitude Encoding: Encode classical values directly into the **amplitudes** of a quantum state. This technique encodes 2^q classical values with q qubits.

Given: Encode a vector of N classical values $\mathbf{x} = [x_1, x_2, \dots, x_N]$ with q qubits

Step 1: Normalize $\mathbf{x}$ into amplitudes of a quantum state $\|\mathbf{x}\|_2^2 = 1$

Step 2: Encode it into $|\mathbf{x}\rangle = \sum_{i=0}^N x|i\rangle$, where i can be single or multi-qubit basis

Variational Quantum Circuits: Encoding Layer

Encoding Layer:

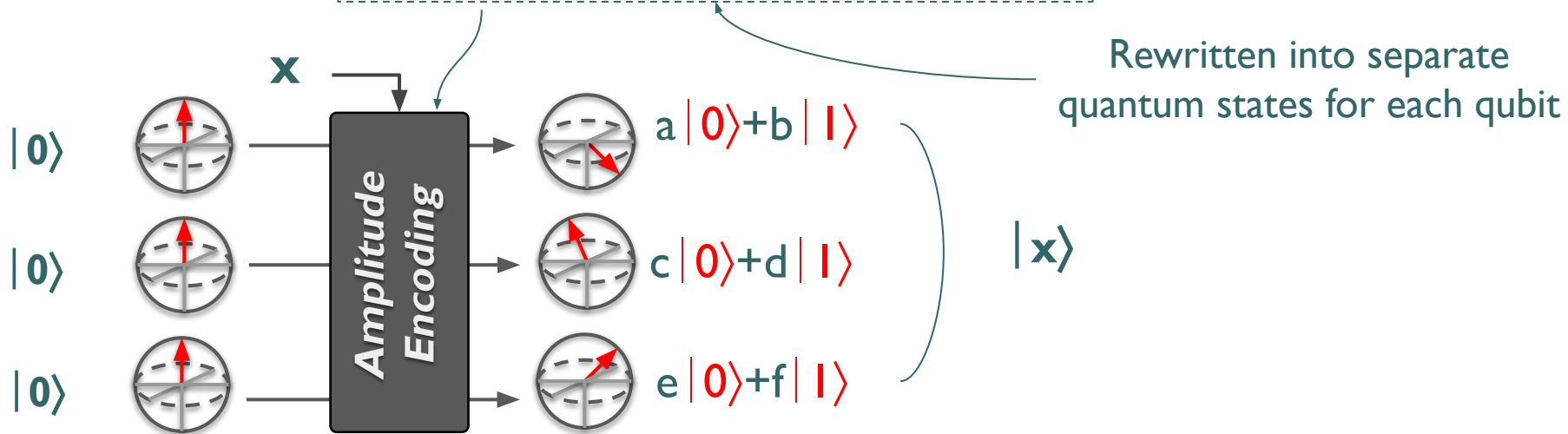
3. Amplitude Encoding:

Example: Encode $\mathbf{x}=[0.12, 0.35, 0.41, 0.22, 0.50, 0.18, 0.44, 0.40]$ with 3 qubits

Step 1: Normalize into amplitudes of quantum state of 3 qubits: $\|\mathbf{x}\|_2^2 = 1$

$$\mathbf{x}=[0.121, 0.352, 0.412, 0.221, 0.503, 0.181, 0.442, 0.402]$$

Step 2: Encode it: $|\mathbf{x}\rangle = 0.121|000\rangle + 0.352|001\rangle + 0.412|010\rangle + 0.221|011\rangle$
 $+ 0.503|100\rangle + 0.181|101\rangle + 0.442|110\rangle + 0.402|111\rangle$
 $= (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) \otimes (e|0\rangle + f|1\rangle)$



Variational Quantum Circuits: Encoding Layer

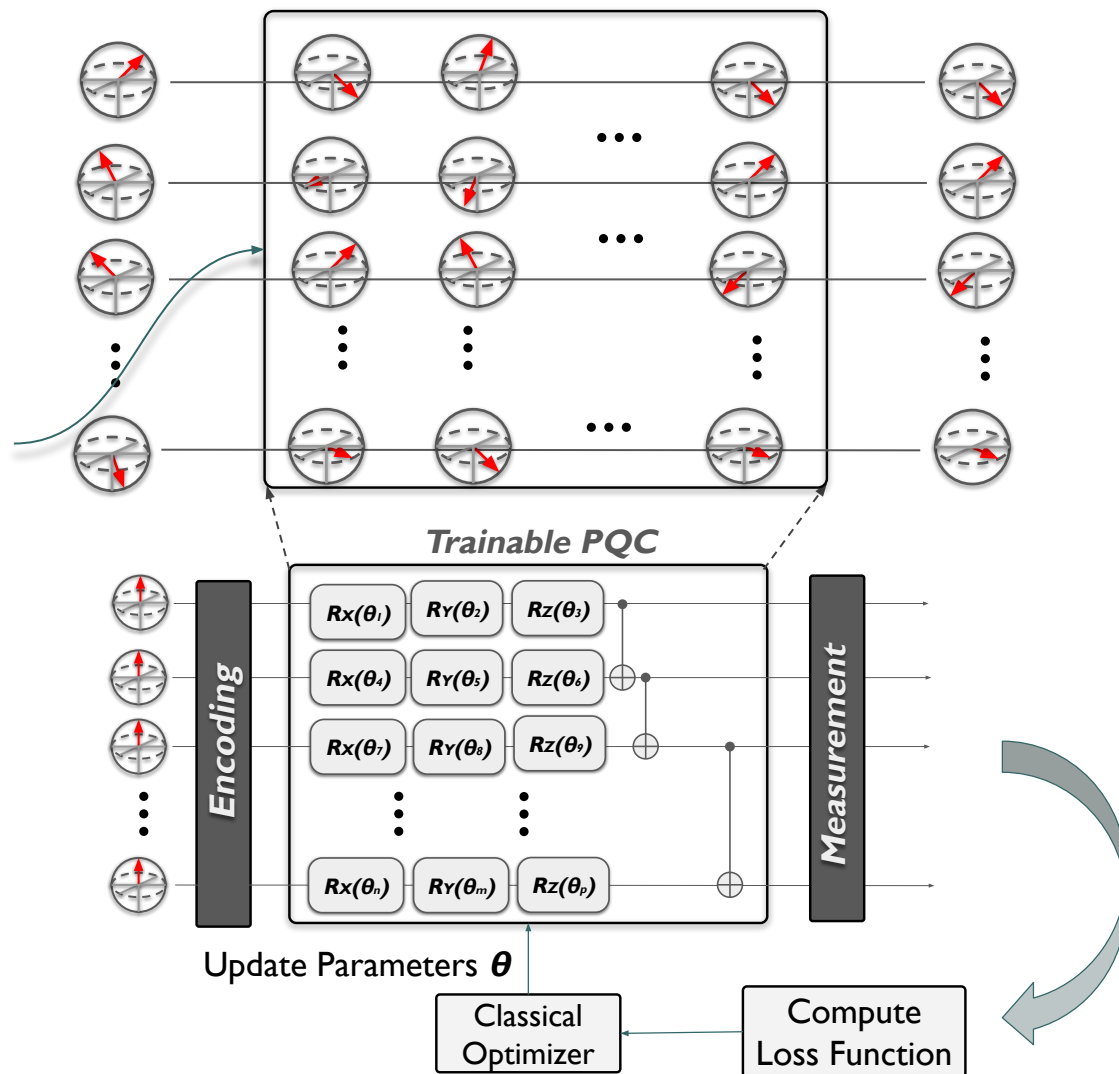
The comparison of 3 Encoding Techniques:

<i>Aspect</i>	<i>Basis Encoding</i>	<i>Angle Encoding</i>	<i>Amplitude Encoding</i>
<i>The number of classical value encoded by q qubits</i>	1 classical value	Few times of q classical values (E.g. 5 qubits can encode 5x3 classical values)	2^q classical values (E.g. 5 qubits encodes 2^5 classical values)
<i>Qubit efficiency</i>	Low	Medium	High
<i>Best use-case</i>	1 scalar value input	Low-dimensional classical input (Dozens)	High-dimensional classical input (Hundreds-Thousands)

Variational Quantum Circuits: PQC Layer

Parameterized Quantum Circuit (PQC) is the quantum version of a trainable quantum gates (like a deep neural network).

Each qubit is rotated over quantum gates

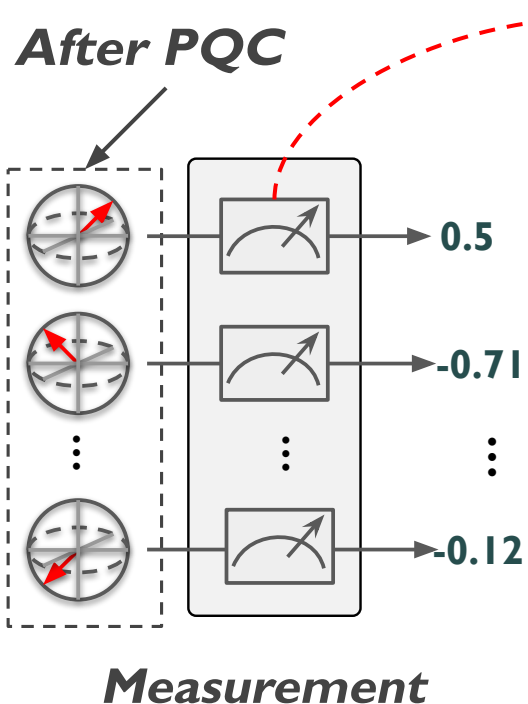


Variational Quantum Circuits: Measurement Layer

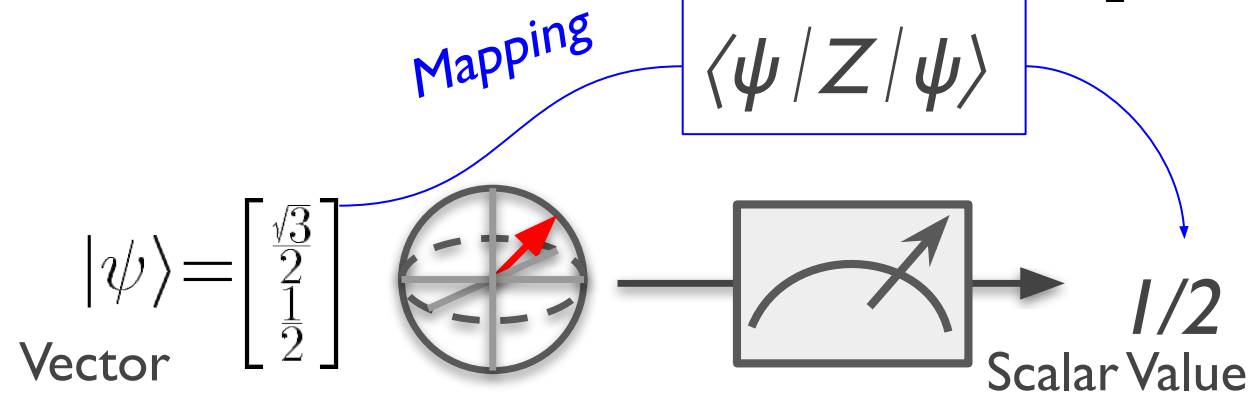
Measurement: is the process of mapping a quantum state to a scalar value.

Example: Pauli-Z measurement, Pauli-Y measurement, Pauli-X measurement

Using matrix operations to map a vector to a scalar value



Pauli-Z measurement using Pauli-Z gate: $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

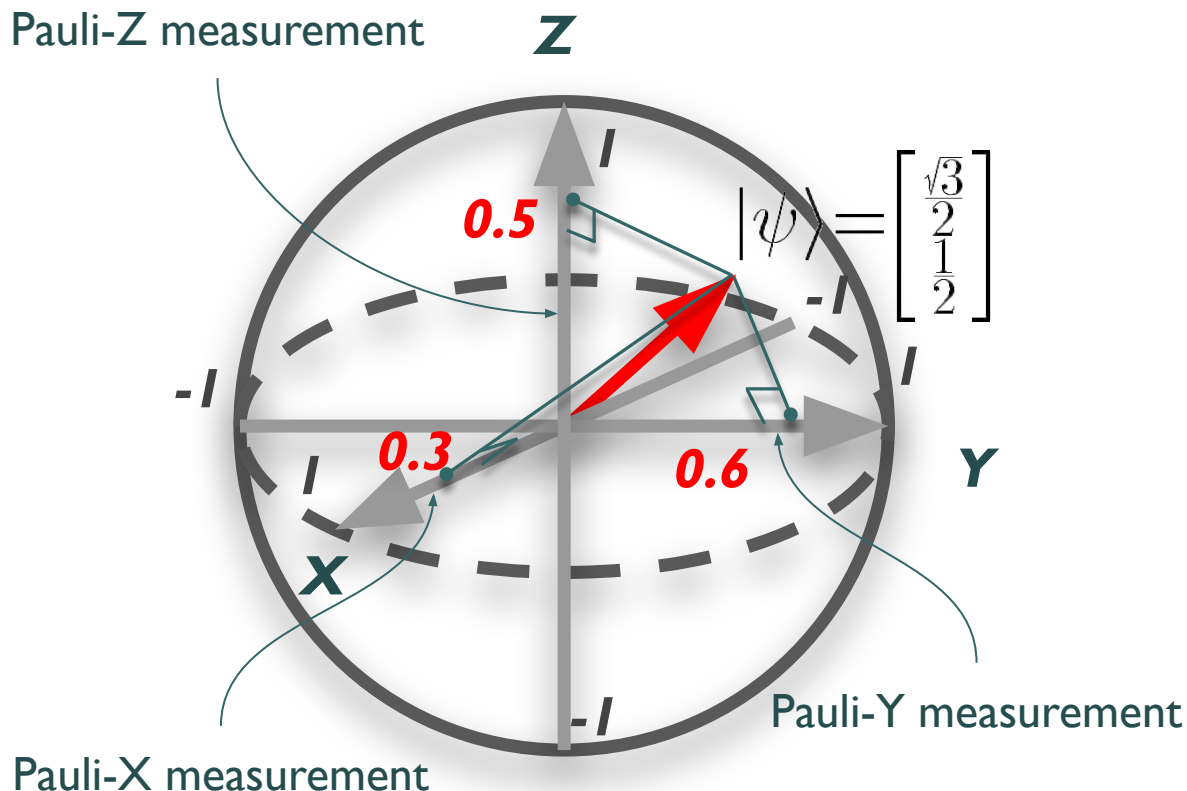


$$\langle \psi | Z | \psi \rangle = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \end{bmatrix} = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

Variational Quantum Circuits: Measurement Layer

Measurement:

Measurement can be understood as the **projection** onto the axis:



QML: Taxonomy

Quantum Machine Learning

```
graph TD; QML[Quantum Machine Learning] --> QSL[Quantum Supervised Learning]; QML --> QUL[Quantum Unsupervised Learning]; QML --> QDL[Quantum Deep Reinforcement Learning];
```

Quantum Supervised
Learning

Quantum Unsupervised
Learning

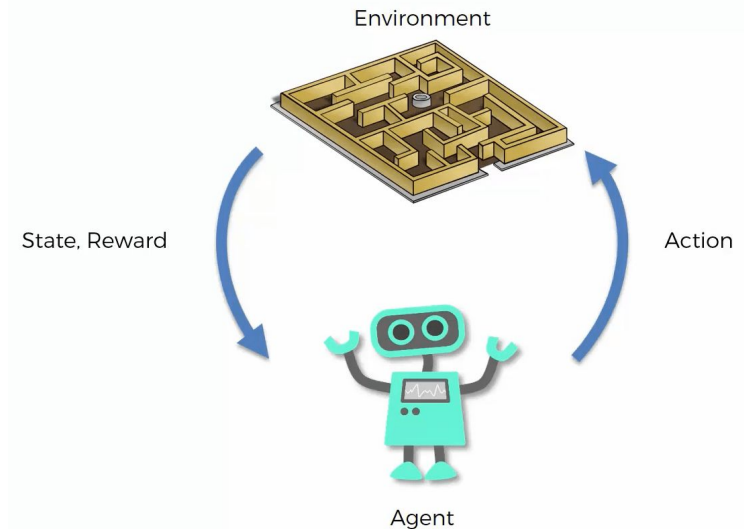
Quantum Deep
Reinforcement Learning



Quantum Deep Reinforcement Learning

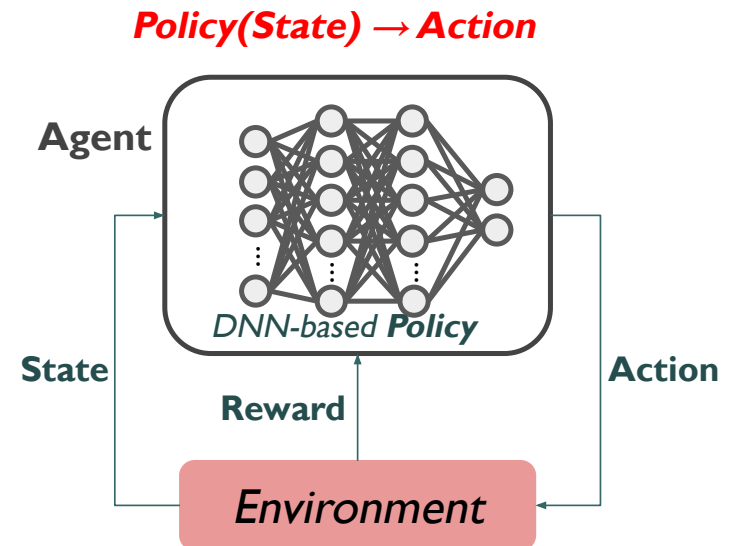
Deep Reinforcement Learning: Definition

Deep Reinforcement Learning (DRL) is the combination of *reinforcement learning* and *deep neural networks*, enabling an agent to learn in an interactive environment by trial and error using feedback from its own experiences



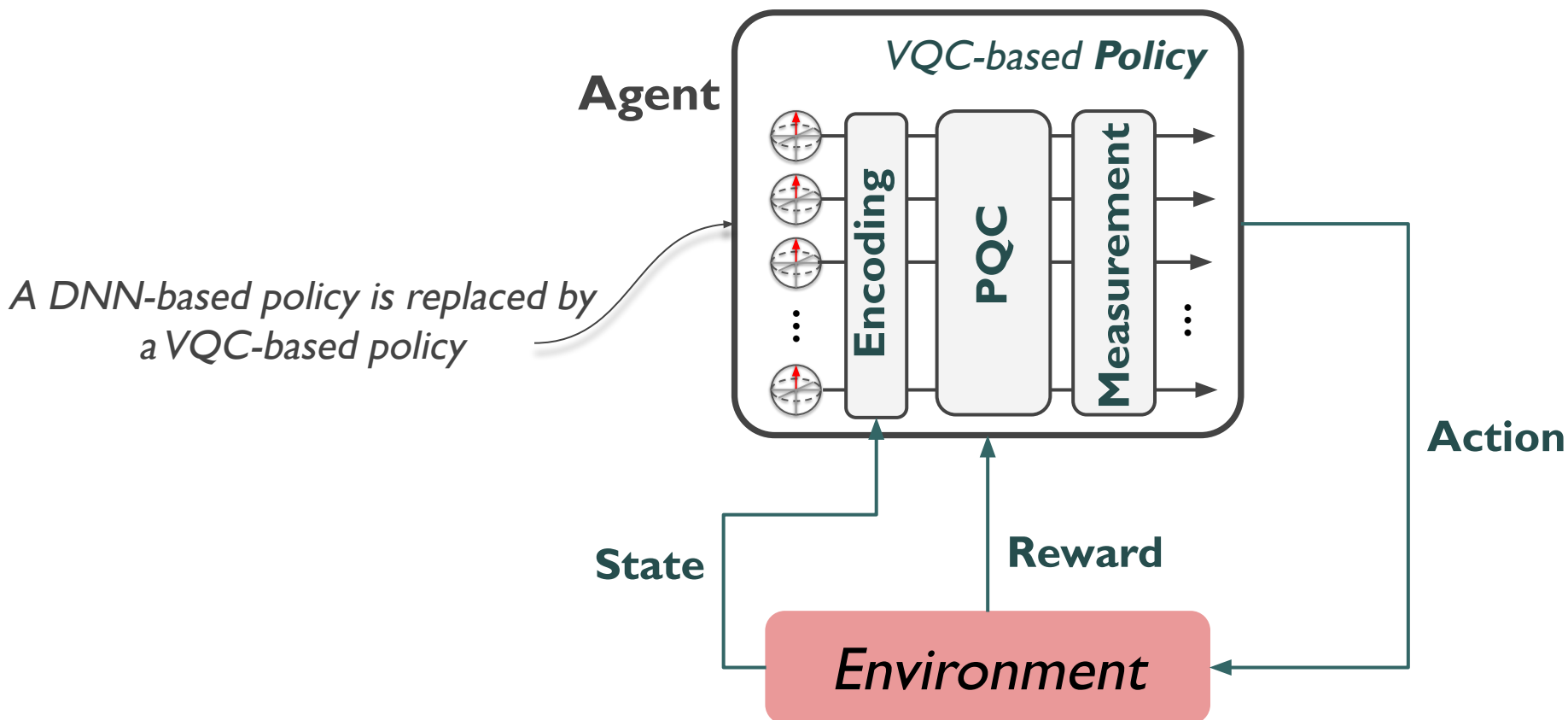
DRL is modeled using a **Markov Decision Process (MDP)**, consisting of:

- **State:** current environment condition
 - **Action:** decision taken by the agent
 - **Reward:** feedback signal from Environment
 - **Policy:** DNN mapping from states to actions
- **Agent:** Maximize the reward

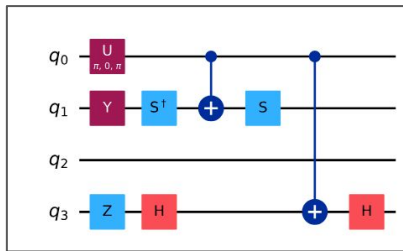


Quantum DRL: Definition

Quantum Deep Reinforcement Learning (QDRL) is a DRL in which **Variational Quantum Circuits (VQCs)** are used as function approximators (replacing classical neural networks) to learn policies or value functions.

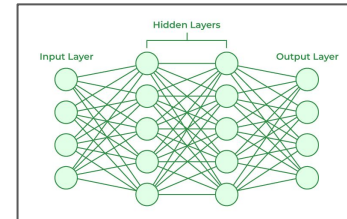


Quantum DRL: QDRL Taxonomy

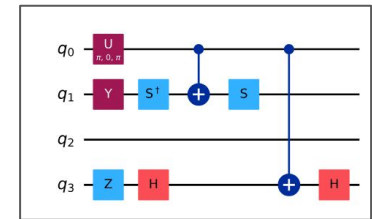


Policy Network uses VQC

Fully QDRL



+



Policy Network uses DNN + VQC

Hybrid QDRL

Quantum DRL

Quantum Q-Learning

VQ-DQN [1]

Quantum Policy Gradient

PQC-based
REINFORCE [2]

Quantum Actor-Critic

PPO-Q [3]

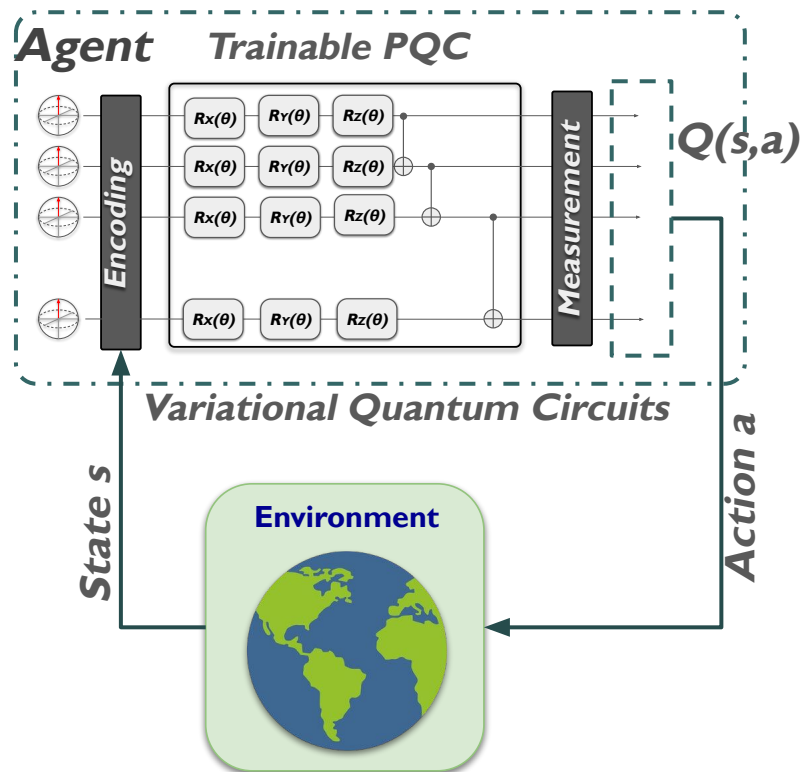
[1] Chen, Samuel Yen-Chi, et al. "Variational quantum circuits for deep reinforcement learning." *IEEE access* vol.8, pp.141007-141024, 2020.

[2] S. Jerbi, et al. "Parametrized quantum policies for reinforcement learning," *Adv. Neural Inf. Process. Syst.*, vol.34, pp.28362–28375, 2021.

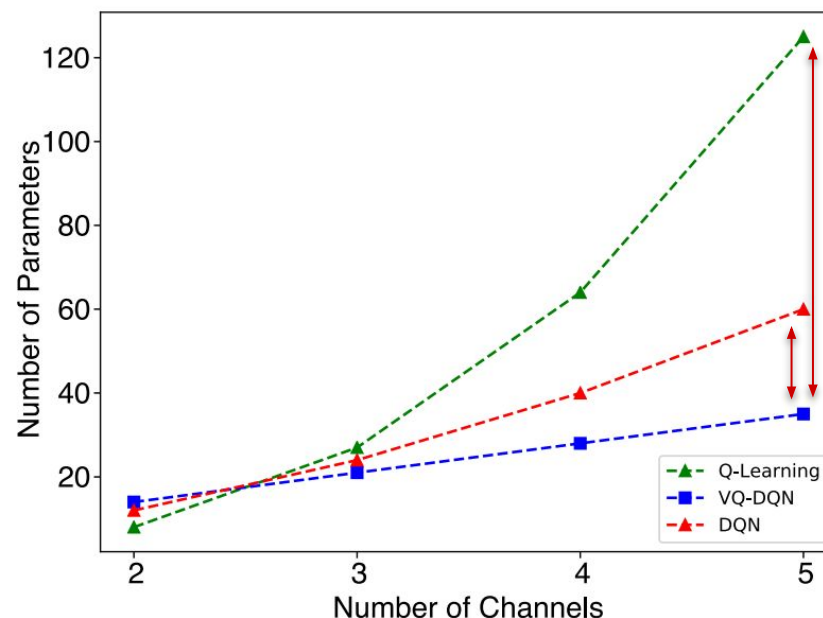
[3] Jin, Yu-Xin, et al. "Ppo-q: Proximal policy optimization with parameterized quantum policies or values." *arXiv preprint arXiv:2501.07085*, 2025.

Quantum Q-Learning: VQ-DQN

The very first QDRL [1] model is Quantum Deep Q-Learning, introduced in 2019.



The number of parameters:



→ QDRL achieves reward convergence comparable to that of classical DRL while significantly reducing trainable parameters.

[1] Chen, Samuel Yen-Chi, et al. "Variational quantum circuits for deep reinforcement learning." *IEEE access* vol.8, pp.141007-141024, 2020

Summary

- A QML circuit (VQC) is architected by 3 key components:
 - (i) Encoding Layer: 3 techniques converting classical value into quantum state
 - (ii) PQC Layer: Trainable Quantum Circuit
 - (iii) Measurement Layer: Convert quantum state into classical value
- A QDRL is a DRL where DNN-based policy is replaced by VQC-based policy
- QDRL Taxonomy:
 - (i) Hybrid QDRL | Fully QDRL
 - (ii) Quantum Q-Learning | Quantum Policy Gradients | Quantum Actor-Critic
- Advantages of QDRL:
 - Significantly reduce the number of parameters while remaining comparable reward convergence
 - For network optimization, some studies show the better reward convergence
- Challenges of QDRL:
 - The limitation of the number of qubits
 - Quantum noise in quantum hardware devices

Thank you for your listening!